

Linear momentum

- Given an object of mass m kg, travelling with velocity v ms^{-1} , we define its **linear momentum** to be mv kg ms^{-1} .
- Linear momentum is a **vector** quantity (since velocity is).

[Formally, we use the term *linear* momentum to distinguish the concept from *angular* momentum, but often we will simply say ‘momentum’ to mean ‘linear momentum’.]

The law of Conservation of Linear Momentum (COLM)

- This principle states that the **total linear momentum** of an isolated system of particles (i.e. which has no external forces acting on it) remains **constant**.
- This principle is derived from Newton’s Laws of Motion.

1. Two spheres, of masses 2 kg and 4 kg, collide and coalesce when they are moving along the same straight line, with speeds 5 ms^{-1} and 3 ms^{-1} respectively.
Find their combined speed after the impact if they were originally moving:
 - a. in the same direction;
 - b. in opposite directions.
2. A sphere of mass 2 kg, travelling at a speed of 3 ms^{-1} , collides and coalesces with a stationary sphere. After the impact, the resulting object has a speed of 1 ms^{-1} .
What was the mass of the stationary sphere?
3. A bullet of mass 0.03 kg is fired horizontally with a velocity of 300 ms^{-1} directly into a stationary wooden block of mass 1.47 kg, which is free to move on a smooth table, and becomes embedded in the block.
 - a. Find the common speed of the bullet and the block after the impact.

4. Spheres A, of mass 2 kg, and B, of mass 3 kg, are traveling in the same direction along the same straight line. They collide when A has a speed of 5 ms^{-1} and B has a speed of 2 ms^{-1} . After the collision B is moving at a speed of 4 ms^{-1} in its original direction. Find the speed of A, and the direction in which it is travelling.
5. A bullet of mass m is fired horizontally with a velocity of $3u$ into a block of wood of mass $7m$ resting on a smooth table. The bullet goes right through the block, and emerges with velocity u . Find the speed of the block after impact.
6. Spheres A, of mass 5 kg, and B, of mass 1 kg, are traveling towards each other along a straight line. They collide when A has a speed of 2 ms^{-1} and B has a speed of 6 ms^{-1} . After the collision, sphere B is moving at 5 ms^{-1} in the opposite direction to its original velocity.
 - a. Find the speed and direction of A.
 - b. Find the loss of kinetic energy caused by the impact.
7. A gun of mass 50 kg fires a bullet of mass 0.04 kg and recoils horizontally. If the bullet leaves the gun with a speed of 300 ms^{-1} , find the speed at which the gun begins to recoil;
8. Three men in a boat each fire five 20 gram rounds astern in rapid succession, each with a speed of 500 ms^{-1} . If the men and the boat together have a mass of 500 kg and the boat was originally stationary, determine the speed with which the boat moves forward after the shots have been fired.
9. A 20 kg shell is travelling horizontally at 400 ms^{-1} when it explodes into two fragments of masses 8 kg and 12 kg. If the 12 kg fragment has a velocity of 700 ms^{-1} in the original direction of motion, determine the velocity of the lighter fragment.
10. [In this question take g to be 10 ms^{-2} .]

A wooden stake of mass 4 kg is to be driven vertically downwards into the ground using a mallet of mass 6 kg. The speed of the mallet just prior to impact is 10 ms^{-1} . After impact the mallet remains in contact with the stake.

- a. Determine the speed with which the stake begins to enter the ground.
 - b. If the ground offers a constant resistance to motion of 1000 N, determine how far the stake will penetrate on each blow.
- 11.

Two particles A and B are moving on a smooth horizontal plane. The mass of A is km , where $2 < k < 3$, and the mass of B is m . The particles are moving along the same straight line, but in opposite directions, and they collide directly. Immediately before they collide the speed of A is $2u$ and the speed of B is $4u$. As a result of the collision the speed of A is halved and its direction of motion is reversed.

(a) Find, in terms of k and u , the speed of B immediately after the collision.

(3)

(b) State whether the direction of motion of B changes as a result of the collision, explaining your answer.

(3)